

Nano-satellites, a fast way to pre-qualify new micro-technology

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Abstract—Micro Systems Technology and Micro-Electronics for space applications can draw great advantages from fast pre-qualification in a real space environment. This is true for technology in general, but probably even more so for these new technologies. Nano-satellites with a mass of 1 to 3 kilograms may provide such a test environment at acceptable cost. In addition they provide an opportunity for close co-operation between industries, institutes and universities, and allow also for active involvement of students in mission, hardware and software design and development during their studies. This paper discusses such an approach and illustrates it by means of a two-year project, Delfi-C³, of the Faculty Aerospace Engineering (AE) and the Faculty Electrical Engineering, Mathematics and Computer Sciences (EEMCS) of Delft University of Technology in the Netherlands. The project is executed in co-operation with Dutch Space and TNO Science & Industry, which supply two of the payloads flown.

I. INTRODUCTION

In recent days much effort is invested in research and development programs for micro-technology for space applications. A major objective of these programs is to reduce size, mass and power consumption of equipment, as these parameters have a direct relation to launch cost and space vehicle development cost. In addition, Micro Systems Technology (MST) and Micro-Electronics (ME) enables new design solutions for space vehicle or instrumentation functions, often with better performance.

Although a standard qualification approach prior to launch is followed, which is generally judged to be a satisfactory guarantee that the equipment will function in a space environment, demonstration of the actual performance during a real space flight is of course more convincing and improves acceptance of the technology for new missions.

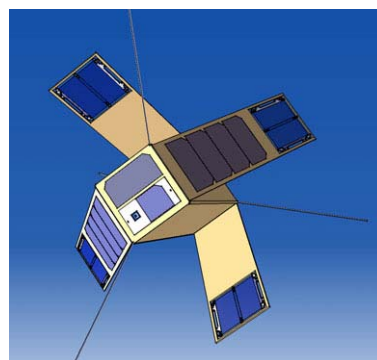
For MST and ME, per definition characterized by relatively small size, low mass and low power demand, nano-satellites with a mass of 1 to 3 kg, a volume of 1 to 3 liters and a typical power of 1 to 3 Watt can be a suitable qualification platform. Since several years a standard is developing for such satellites. Originally initiated in 2000 by California Polytechnic State University San Luis Obispo (Cal Poly) and Stanford University's Space Systems Development Lab, the CubeSat concept has been developed leading to a first launch of a number of university satellites in June 2003 [1]. Since then the concept has been adopted by industry, so that standard CubeSat kits and subsystems are available, that can be used as a basis for a complete satellite [2]. Launch services are brokered by e.g. Cal Poly and University of Toronto Institute for Aerospace Studies for as few as \$40,000 for a satellite of 1

kg and a volume of 10x10x10 cm.

Although originally intended to offer educational institutions an affordable means to access space, the same set-up could be used to pre-qualify new MST and ME based systems in space. That approach has been chosen for Delfi-C³, a two-year project of the Faculty Aerospace Engineering (AE) and the Faculty Electrical Engineering, Mathematics and Computer Sciences (EEMCS) of Delft University of Technology in the Netherlands, executed in co-operation with Dutch Space and TNO Science & Industry.

II. DELFI-C³

The idea for Delfi-C³ originated from the desire of Dutch Space to have an early flight opportunity (end of 2006) for thin film solar cells. This new solar cell concept is aiming to reduce the mass of solar panels drastically. The solar cells have a substrate of Titanium foil, are only 50 μm thick and are almost individually suspended in a frame to form the solar panel. Main objective of the space mission is to measure the U-I curves of the solar cells as a function of operating temperature and of angle of incidence of the solar radiation.



Although the solar cell itself is not a MST and ME based system, flying them would allow pre-qualifying such technology. Candidate applications were collected from the MISAT program [3], [4], a five-year program starting in 2005 to develop nano-satellites that demonstrate a wide range of MST and ME applications for use in space vehicles and instrumentation. Applications selected are an autonomous sun sensor with its own power supply and a wireless data connection (TNO Science and Industry), which was anyhow needed to measure the position of the sun vector, and a high efficiency transceiver (Faculty EEMCS of the TU Delft). The transceiver is used to receive commands from the Delfi ground station and transmit measurement and housekeeping data back to the ground station, but that is not its primary mission. The satellite will make use of the amateur radio

frequency bands, so the transceiver will be compatible with the AMSAT standards and operate as a transponder that can be used by radio amateurs. In return the radio amateurs are asked to collect the Delfi-C³ data and send them to the Delfi ground station in Delft. In this way, data can be collected from the satellite even when it is not visible for the Delfi ground station. A detailed description of Delfi-C³ is given in [5].

III. MICRO-TECHNOLOGY FLOWN

A. Autonomous Wireless Sun Sensor

Delfi-C³ houses two analogue Sun Sensors (located at opposite sides of the satellite) that will be fully autonomous and wireless. With the typical internal volume of a nano-satellite already being quite limited, reducing wiring is an important challenge. To this end, the sensors will have a half-sized solar cell as their own power supply, making them independent of the satellite's electrical power system. This autonomy is accompanied by a wireless Radio Frequency link. This link will be either an adapted commercial off-the-shelf 868 MHz band or a custom designed Ultra-Wide Band connection, depending on data rates and technology readiness. Although wireless data communication on board such small satellites might seem a bit superfluous in itself, implementing this wireless technology adds to modularity and results in a flexible "plug and play" system. The autonomous sensor unit, in this case measuring approximately 80x40x20 mm, is a predecessor of an even smaller digital version that can be mounted on e.g. a solar panel sacrificing just a single solar cell.

The in orbit experiment will concentrate on demonstrating the feasibility of the wireless link (immunity for disturbances; no interference with other equipment) and the operation of the sun sensor under strongly variable power supply.

B. High Efficiency Transceiver

One of the MST/ME experiments on board will be the high efficiency transceiver. Nano-satellites (for MISAT) have not much room for solar panels and batteries, so very limited power is available. Often the power system cannot provide the transmission power required by the antenna, especially because of the limited efficiency of the power amplifier (PA) in the transceiver. Methods exist to optimize the efficiency of the PA, but they tend to limit the operating bandwidth of the amplifier and the choice of modulation scheme. This makes application of these PA's in nano-satellites problematic. Nano-satellites may perform a mission together in close cooperation. This implies that apart from communication to a ground station at various frequencies and data rates and apart from internal wireless communication with the on-board payloads, there will also be inter-satellite communication at variable frequencies and data rates. So there will be many frequencies, bandwidths and modulation schemes in use in one nano-satellite formation. Restrictions on weight, volume and power consumption make it impossible to fly separate

radios for each of the wireless links. One transceiver should be able to accommodate for all wireless communication. A crucial part in this is the availability of a wide-band, highly efficient and very linear PA. In this mission a PA will be tested that makes use over-all double loop negative feedback, using integrated transformers in the feedback network and in the frequency compensation network. It has been shown that integrated transformers work very well in a wide band when used as feedback element [6]. Since negative feedback is a very powerful means to linearize an amplifier, the more non-linearity of the active part-especially the power stage- can be tolerated. When more non-linearity can be tolerated, it is possible to increase the efficiency of the power stage. Several experiments have shown that *both* a higher efficiency and a higher linearity can be obtained using this technique. Space-qualification of this technique, where integrated transformers are optimized for wide-band use in feedback amplifiers is crucial. The circuit diagram is shown in fig.2. Table 1 shows the improvement that has been achieved in a first experiment comparing amplifiers with identical active parts (and efficiency), one with and one without feedback.

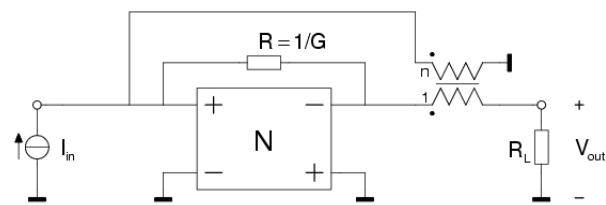


Figure.2 A transformer based double loop negative feedback amplifier.

IM3	0.9GHz	2.0GHz
FB	-46dBc	-43dBc
NO FB	-29dBc	-33dBc
Improvement	17dB	10dB

Table 1 Comparison between amplifiers with and without feedback.

REFERENCES

- [1] <http://cubesat.calpoly.edu>
- [2] <http://www.pumpkininc.com>
- [3] W. Jongkind, "The Dutch MST program MicroNed and its cluster MISAT", (submitted for publication at the ICMENS 2005).
- [4] F. van Keulen et.al. "MicroNed project plan", *Internal Document*. Delft, Febr. 2005.
- [5] A.R. Bonnema, et al., "Delfi-C3: a student nano-satellite as a test-bed for thin film solar cells and wireless onboard communication", (accepted for publication at the 2nd International Conference on Recent Advances in Space Technologies, Istanbul, Turkey, June 2005)
- [6] Koen van Hartingsveldt, Michiel Kouwenhoven, Chris Verhoeven, "HF low noise amplifiers with integrated transformer feedback", *ISCAS 2002*, USA